UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Saturday 09 June 2018 9:00 to 12:00

Paper PHA/WP

PHARMACOLOGY (Written Practical)

*Answer **all three** questions.*

*Start **each** question on a new sheet*

*Tie the answer to each question separately, with your **candidate number** on each cover sheet*

Calculator – students are permitted to bring an approved calculator

STATIONARY REQUIREMENTS

Metric graph paper x 4 sheets
5-cycle semi-log graph paper x 4 sheets
Script paper
Cover sheets

SPECIAL REQUIREMENTS

Approved calculator

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the invigilator

Question 1

The voltage-gated Na^+ channel (Na_V) family of ion channels consists of nine different proteins, $Na_V1.1$ – $Na_V1.9$, which are differentially expressed throughout the body. The expression of $Na_V1.7$ is largely restricted to the sensory nervous system and as such many pharmaceutical companies have focused on developing specific $Na_V1.7$ inhibitors that could be used to relieve pain without causing side effects associated with inhibiting the activity of other Na_V subunits.

The Wundermittel pharmaceutical company have recently identified two new compounds that they believe to be inhibitors of $Na_V1.7$, E4 and R2A. To determine the action of E4 and R2A on the function of $Na_V1.7$, human embryonic kidney (HEK) cells expressing human $Na_V1.7$ have been loaded with a fluorescent dye, the fluorescence of which increases during membrane depolarisation. Veratridine causes depolarisation of cells by abolishing the inactivation of Na_V subunits. The effects of E4 and R2A on veratridine-induced increases in fluorescence were measured. The results are given in Table 1. You can assume that the HEK cells used express no other Na_V subunits.

Table 1: *Fluorescence changes in HEK cells expressing $Na_V1.7$ following application of veratridine*

	Fluorescence (%)		
[veratridine] (M)	Absence of antagonist	E4 (10 nM)	R2A (10 nM)
0.0000001	3	3	2
0.0000005	18	7	14
0.000005	45	28	33
0.00003	71	50	57
0.0005	93	76	84
0.01	100	100	100

a) What are the affinity constants of E4 and R2A for $Na_V1.7$? (5 marks)

b) How could the reliability of the affinity constants be improved? (1 mark)

A limitation of the fluorescence based method described above is that it measures membrane depolarisation, not $Na_V1.7$ activity. Therefore, whole-cell patch-clamp electrophysiology was used to directly measure $Na_V1.7$-mediated currents by depolarising cells from a variety of different resting membrane potentials. It was found that E4 had an IC_{50} of 8.67 µM for currents evoked from a resting membrane potential of -100 mV, but an IC_{50} of 0.37 µM for currents evoked from a resting membrane potential of -30 mV. By contrast, R2A showed negligible inhibition of currents when evoked from a resting membrane potential of -100 mV, but had an IC_{50} of 0.55 µM for inhibition of currents evoked from a resting membrane potential of -30 mV.

c) Explain these results. (2 marks)

[Question 1 continues on the NEXT PAGE]

To further characterise the properties of R2A, a tritiated version was made and a radioligand binding assay conducted using 0.05 mg of rat sensory neurone membranes. Membrane samples were incubated with various concentrations of [^{3}H]R2A in the absence and presence of a high concentration (10 µM) of unlabelled R2A. Samples were first filtered through glass fibre filters and then rapidly washed with ice- cold buffer solution. The radioactivity associated with the filters was then counted by liquid scintillation counting. The results obtained are given in Table 2.

Table 2: *Binding of [^{3}H]R2A to rat sensory neurone membranes*

	Binding (d.p.m.)	
[^{3}H]R2A nM	Radioligand only	Radioligand plus 10 µM R2A
0.07	200	150
1	810	220
2	1260	270
5	2050	380
10	2650	500
25	3620	1000
45	5000	2200

The specific activity of [^{3}H]R2A was 0.03 Ci/mmol (1 Ci = 2.22 x 10^{12} d.p.m.).

d) Determine the affinity constant and the binding capacity (B_{max}) for R2A on rat sensory neurone membranes. (5 marks)

e) Comment on the affinity constant of R2A calculated using the functional, fluorescence assay conducted in HEK cells in comparison to the value calculated using rat sensory neurone membranes. (1 mark)

A similar assay was carried out using rat sensory neurone membranes with tritiated and non-tritiated versions of E4. The binding capacity with E4 was determined to be 14 nmol/mg.

f) How might the difference between this B_{max} and that of R2A be explained? (1 mark)

[Question 1 continues on the NEXT PAGE]

In a final set of experiments, the fluorescence assay was repeated in HEK cells expressing $Na_V1.5$, rather than $Na_V1.7$. The EC_{50} of veratridine was calculated as 110 µM in the presence of E4 and 5 µM in the presence of R2A.

g) What can be concluded from these experiments and what do they suggest about the clinical potential of E4 and R2A2? (3 marks)

h) What further experiments could be conducted to support your conclusions? (2 marks)

End of Question 1

[Question 2 starts on the NEXT PAGE]

Question 2

Metformin is a first-line medication for type 2 diabetes mellitus. Metformin is a hydrophilic base with a pKa of 12.4 and predominantly exists as a cation at physiological pH. Organic cation transporters (OCT) are members of the solute carrier (SLC) 22A superfamily and are the major transporters for movement of many cationic drugs across plasma membranes. As such, they have been implicated in the pharmacokinetics of metformin. To study this, 0.25 mg metformin was injected into the tail vein of wild-type mice (WT), transgenic mice in which *Slc22a1*, the gene encoding OCT1, had been deleted (*Slc22a1*$^{-/-}$), or transgenic mice in which the genes encoding both OCT1 and OCT2 had been deleted (*Slc22a1*$^{-/-}$.*Slc22a2*$^{-/-}$). Plasma samples were taken at regular intervals. Mice were housed in a metabolic cage to allow urine collection.

All procedures involving protected animals were conducted in accordance with the United Kingdom Animal (Scientific Procedures) Act 1986 Amendment Regulations 2012 and were approved by the local Animal Welfare Ethical Review Body. The mice in the experiments were age-matched. Only male mice were used. On average, the mice weighed 30 g.

The mean plasma concentrations of metformin measured in the three strains are shown in Table 1.

Table 1: *Mean plasma concentrations of metformin in the three genotypes of mice.*

Time (min)	Metformin plasma concentration (µg/ml)		
	WT	*Slc22a1*$^{-/-}$	*Slc22a1*$^{-/-}$.*Slc22a2*$^{-/-}$
10	4.01	5.31	16.73
20	2.57	2.05	11.91
30	1.65	0.79	8.48
45	0.85	0.19	5.10
60	0.44	0.05	3.06

a) Calculate the clearance and terminal phase volume of distribution of metformin in each of the three mouse genotypes. (6 marks)

b) Explain what role organic cation transporters play in the distribution of metformin in mice. (3 marks)

At the end of the experiments, the excretion of metformin into the urine was measured. The percentage of administrated metformin found unchanged in urine is shown in Table 2.

Table 2: *Urinary excretion of unchanged metformin in the three strains of mice.*

	WT	*Slc22a1*$^{-/-}$	*Slc22a1*$^{-/-}$.*Slc22a2*$^{-/-}$
% of dose in urine	95	95	75

In all genotypes, the GFR was estimated to be 0.28 ml/min.

c) How does urinary clearance vary between the three genotypes? What does this show about the role of organic cation transporters in elimination of metformin in mice? (4 marks)

[Question 2 continues on the NEXT PAGE]

To investigate the potential clinical relevance of these findings in humans, healthy volunteers were given 0.25 g metformin by oral administration. A week later, the same volunteers were given 0.25 g metformin together with cimetidine, a histamine H2 receptor antagonist. Cimetidine is known to block OCT2. Plasma concentrations and urinary excretion of metformin were regularly measured. The results of these experiments are shown in Table 3.

Table 3: *The effect of co-administration of cimetidine on the pharmacokinetics of metformin.*

	Metformin	Metformin + cimetidine
C_{max} (µg/ml)	0.59	1.02
Fractional bioavailability	0.55	0.55
AUC (µg.h/ml)	3.95	7.79
Renal clearance, CL_{RENAL} (ml/min)	527	278

d) To what extent do these data support your interpretation of the mouse experiments? (2 marks)

e) If a patient were given 0.25 g metformin as a daily dose, what would be the average plasma concentration at steady state? What would be the average plasma concentration at steady state if the patient were also taking cimetidine? (5 marks)

End of Question 2

[Question 3 starts on the NEXT PAGE]

Question 3

Bromodomain-containing proteins (BRDs) act as epigenetic 'readers' to regulate gene transcription by recruiting transcriptional machinery to promoters modified by histone acetylation. Several BRDs exist (BRD1-4) and are considered attractive therapeutic targets in cancer. Small molecule inhibitors currently in clinical trials include JQ1, a potent and selective BRD inhibitor that has helped demonstrate the essential role of BRDs in controlling cell growth via expression of target genes such as c-myc. JQ1 binds BRDs to block their interaction with acetylated lysines in histone tails.

A promising new strategy in drug design consists of inducing degradation of targets using so-called 'targeted protein degraders'. These bifunctional molecules link small molecule ligands for target proteins to small molecules that bind specific components of the cellular protein degradation machinery, effectively 'hijacking' the machinery to destroy the target.

A drug discovery programme to design targeted protein degraders of BRDs tested a new BRD ligand, PH3, linked to thalidomide via a variable linker to generate a number of new compounds for preclinical development, including two that showed promising properties in initial tests, Compounds A & B (Figure 1).

Figure 1: *Schematic of drugs targeting BRDs used in this study*

The binding properties of compounds to the bromodomain of BRD2 were tested by competitive binding assays using a fluorescently-labelled acetylated histone tail peptide. Compounds were then tested for their effect on growth of two human leukaemia cell lines (CRL-4253 and TIB-592). These data are summarized in Table 1.

Table 1: *Pharmacology of drugs targeting BRDs*

	IC_{50} (nM) for binding to BRD2	linker	IC_{50} (nM) for growth inhibition	
			CRL-4253 cells	TIB-592 cells
JQ1	36.0 ± 5.1	-	143 ± 22	151 ± 18
PH3	3.1 ± 0.7	-	24.1 ± 5.3	25.6 ± 1.9
Compound A	4.6 ± 0.9	$CH_2CH_2CH_2CH_2CH_2$	0.051 ± 0.018	2.2 ± 0.2
Compound B	3.3 ± 0.7	$(CH_2)_4NHCOCH_2NH$	4.3 ± 3.1	45.5 ± 14.1

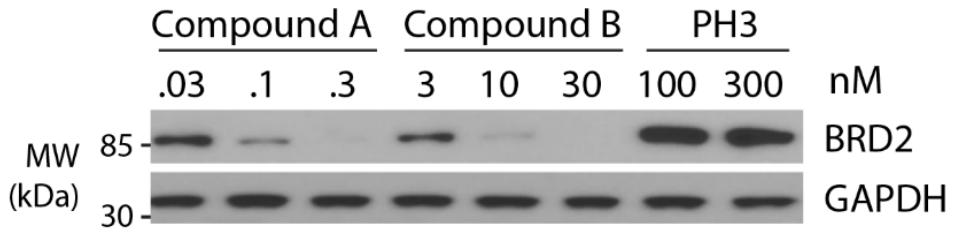
a) Explain the data shown in Table 1 as fully as possible. (5 marks)

[Question 3 continues on the NEXT PAGE]

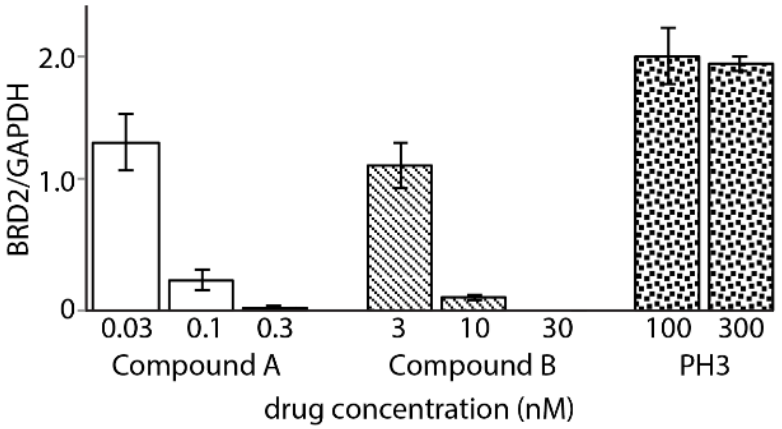
Experiments were carried out to investigate the molecular basis for growth inhibition by drugs targeting BRDs. CRL-4253 cells were treated for 3 hours with drugs at the concentrations indicated in Figures 2 – 4, then lysed to make cell extracts. 20 μg of each extract was loaded onto gels for electrophoresis and preparation of Western blots (immunoblots) that were probed for expression of different cellular proteins using antibodies indicated to the right of panels. Cleavage of PARP (Poly-ADP Ribose Polymerase) by activated caspases to generate a shorter fragment (PARP*) is an early marker of apoptosis. Each experiment was repeated three times, with a representative image shown for each experiment (Figures 2 – 4). Cellular protein levels were quantified by measuring the intensity of bands revealed by immunoblot, using an abundant, constitutively-expressed protein (such as GAPDH) as a loading control (Figure 2).

Figure 2: *Testing cellular levels of BRD2. The immunoblot shown in panel* ***A*** *was quantified as described in the text to obtain measurement of relative BRD2 levels (panel* ***B****).*

A



B



b) Comment on this method for protein quantification. What assumptions are made? (2 marks)

[Question 3 continues on the NEXT PAGE]

Figure 3: *Testing specificity of a targeted protein degrader. Cells were pretreated for 2 hours with doses of PH3 as shown, before 3 hours treatment with 1 nM Compound A or DMSO (solvent control). Immunoblots were examined with antibodies against different BRD family members, BRD2-4, or GAPDH.*

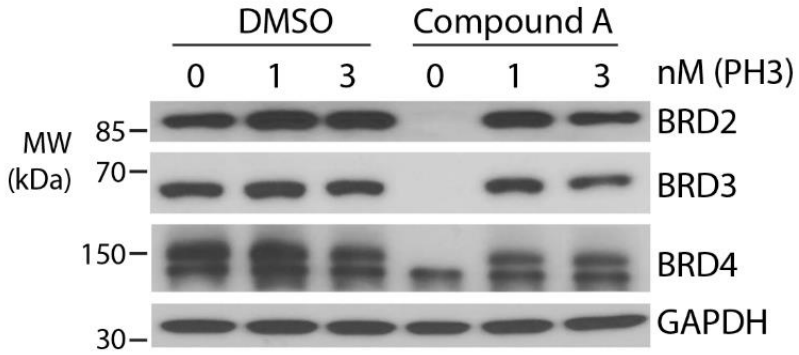
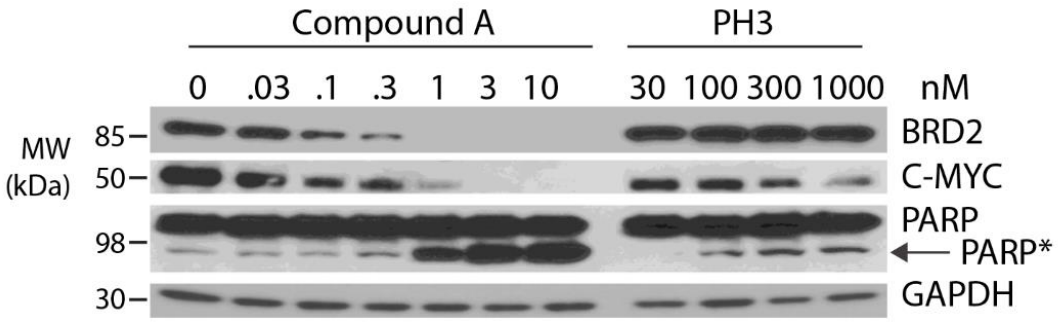


Figure 4: *Examining apoptotic markers in response to drug treatment. Immunoblots were additionally examined using antibodies specific to c-myc and to PARP.*



c) Discuss what is shown in each of Figures 2 – 4. (6 marks)

The fate of cells treated with drugs targeting BRDs was further investigated. Cells were treated for 24 hours with drug doses indicated in Figures 5 and 6, then stained with propidium iodide to measure DNA content (Figure 5). PARP* formation was used to measure apoptosis (Figure 6).

[Question 3 continues on the NEXT PAGE]

Figure 5: *Cell cycle phase distribution of CRL-4253 populations. Propidium iodide fluorescence was measured in 1 x 10^6 individual cells using a flow cytometer. The number of cells assigned to each cell cycle phase is expressed as a percentage of the total number.*

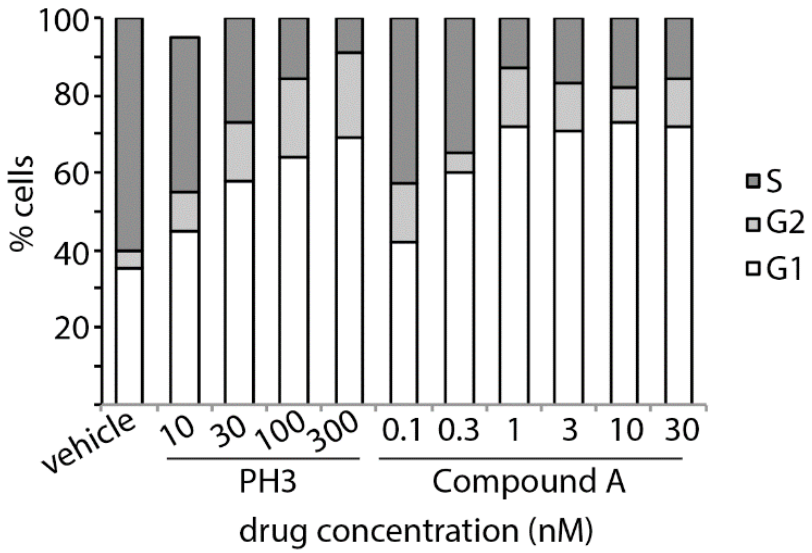
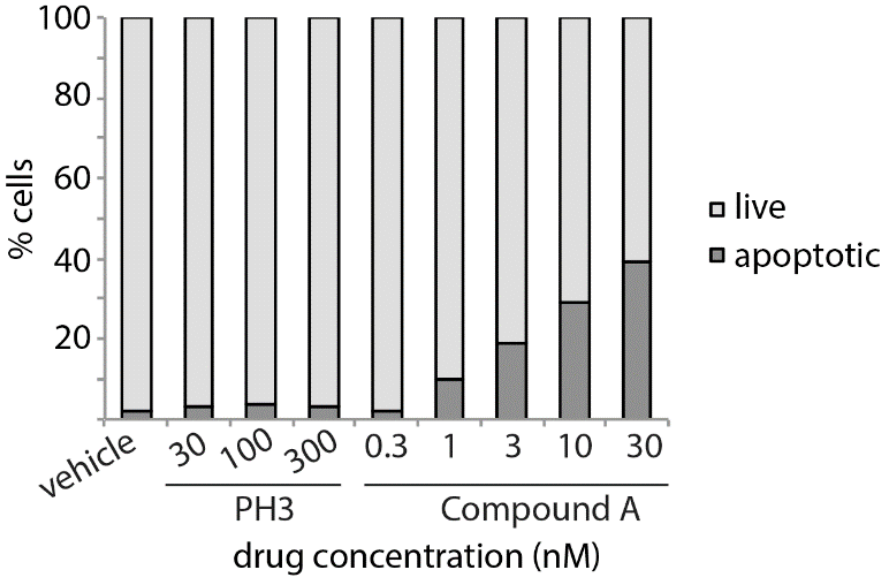


Figure 6: *Viability of CRL-4253 cells. PARP* levels were measured in 10^6 individual cells using a fluorescently-labelled antibody specific for PARP*. The number of cells assigned to alternative cell fates is expressed as a percentage of the total number.*

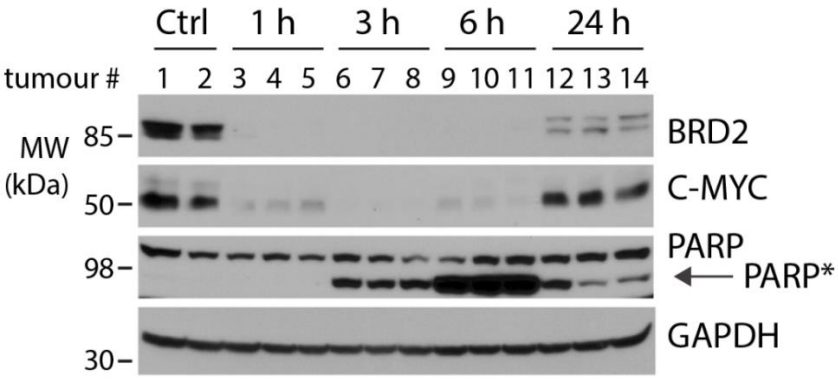


d) Discuss the differences in cell fate following treatment with inhibitor (PH3) or degrader (Compound A). (4 marks)

[Question 3 continues on the NEXT PAGE]

Pharmacodynamic analysis *in vivo* was carried out using a mouse xenograft model, in which tumour tissues derived from CRL-4253 cells were injected into mice to cause tumour growth and then harvested at the times indicated following a single intravenous dose of Compound A at 5 mg kg^{-1} (Figure 7). Control tumours (from untreated mice) were harvested at 6 hours.

Figure 7: *Protein expression in tumours harvested at various times after treatment with Compound A. Extracts prepared from tumour tissue were examined by immunoblot for levels of BRD2 and markers of apoptosis. (Ctrl = control)*



e) Comment on this experimental design and suggest additional experiments that would support preclinical development of Compound A. (3 marks)

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